A Migrating Bird's Wings Trade Speed for Sharp Turns

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The European nightjar is a bird that hunts insects at night. Every year, it **migrates** thousands of miles between Europe and southern Africa. You might expect a bird that travels so far to have wings built for fast, efficient flying. But scientists have found that the nightjar's wings make a trade-off.

Researchers at Lund University studied nightjars in a **wind tunnel**. They recorded how the birds moved at different speeds and measured the patterns of air left behind their wings.

The nightjar has broad wing tips. These likely help it make sharp turns while flying slowly to catch insects. But at higher speeds, the broad tips become a disadvantage, and the wings make **lift** less efficiently. **Lift** is the force that helps keep a bird in the air.

"Instead, the nightjar's wings appear to be a **compromise** between several different demands," says Christoffer Johansson, a scientist at Lund University.

The team also found a surprise. Even as the nightjar raises its wings, during the part of the wingbeat called the **upstroke**, it keeps creating **thrust**. **Thrust** is the force that moves it forward. This is unusual, because an active **upstroke** like this has mainly been linked to bats and insects, not birds.

The finding suggests that very different animals may have found similar ways to solve the problems of flying.

The discovery could also help engineers. Drones and robots with flapping wings may face limits on how much their wings can fold during each beat. Learning how nightjars push forward on the **upstroke** could help such machines fly better.

Next, the researchers want to find out whether other birds and flying animals use the same trick, or whether it is special to the nightjar.

Word Treasure

migrates -- Travels a long way each year between two regions.

wind tunnel -- A tube where air is blown past an object so scientists can study how air moves around it.

lift -- The upward force that helps keep something in the air.

thrust -- The force that pushes something forward.

upstroke -- The part of a wingbeat when the wings move upward.

compromise -- A balance between different needs where each gives up a little.

Background you need

Many birds migrate, flying long distances between seasonal homes to find food and good weather.

Flight depends on forces: lift holds a bird up, and thrust moves it forward.

Scientists use wind tunnels, where air is blown past a flying animal, to study how wings move air.

A bird's wing shape often reflects its lifestyle, such as hunting, soaring or traveling far.

Break it down

LEAD: The nightjar's wings trade high-speed efficiency for agility
METHOD: Lund University researchers recorded nightjars in a wind tunnel and measured air patterns
KEY FINDING: Broad wing tips help slow, sharp turns but make lift less efficiently at high speed
+- WHO: Christoffer Johansson, Lund University
+- QUOTE: The wings appear to be a compromise between several different demands
SURPRISE: The nightjar creates thrust even on the upstroke, as bats and insects do
APPLICATION: Could help flapping-wing drones and robots
OPEN QUESTION: Do other birds and flying animals use the same trick?

Why it matters

The study shows that evolution often balances competing needs instead of perfecting one, and nature's solutions can inspire new flying machines.

Test what you remember

1. Where does the European nightjar migrate each year?

- (A) Between Europe and southern Africa
- (B) Between North and South America
- (C) Between Canada and Mexico
- (D) Between Asia and Australia

2. Where did researchers study the nightjars?

- (A) In a zoo
- (B) In a rainforest
- (C) In outer space
- (D) In a wind tunnel

3. What do the nightjar's broad wing tips likely help it do?

- (A) Dig nests
- (B) Swim underwater
- (C) Fly faster than jets
- (D) Make sharp turns while flying slowly

4. What is lift?

- (A) A way of folding wings
- (B) The force that slows a bird down
- (C) A kind of insect
- (D) The force that helps keep a bird in the air

5. Which animals had scientists mainly linked to an active upstroke?

- (A) Fish and frogs
- (B) Bats and insects
- (C) Snakes and lizards
- (D) Penguins and ostriches

6. How could the discovery help engineers?

- (A) By designing faster boats
- (B) By helping flapping-wing drones and robots fly better
- (C) By making cars quieter
- (D) By building taller bridges

Answer key (try the quiz first!): 1: A 2: D 3: D 4: D 5: B 6: B

Questions to think about

1. Christoffer Johansson, Lund University: Says the nightjar's wings are a compromise between several different demands.

MY THOUGHTS (at least 20 words):